HARP v4.1 — Mathematical Supplement

Statistical machinery and implementation-level definitions

Scope and implementation authority

This supplement documents the mathematical operations implemented in HARP v4.1. It is intended as the technical companion to *Interpreting HARP v4.1 Graphical Output*. The graphical guide addresses what HARP outputs show and how they should be interpreted; this supplement addresses the narrower question: *what, mathematically, does HARP v4.1 calculate?*

The equations below were derived directly from the implementations in `trajectory.py`, `deepcoil.py`, `stats.py`, and `analysis.py`. `plotting.py` was additionally inspected to verify that graphical presentation introduces no further inferential transformation. The mathematics therefore describes HARP v4.1 as implemented, rather than a generalized or conventional statistical formulation.

1 Notation and indexing

HARP operates on an MD-derived contact signal indexed against a DeepCoil2-anchored seven-state heptad coordinate. The principal notation is:

- $t = 1, \dots, F$: analysed trajectory frames;
- i: MyhT residue index;
- j: MG residue index;
- p_i: one-based MyhT sequence position;
- $k = 0, \dots, 6$: heptad phase index corresponding to a, b, c, d, e, f, g;
- w_{tji}: MG residue j–MyhT residue i contact score in frame t;
- x_{ti}: per-frame MyhT residue contact signal;
- x_i: time-averaged MyhT residue contact profile;
- q_i: HARP heptad phase index assigned to residue i;
- S_k: summed contact signal assigned to phase k;
- n_k: number of residues assigned to phase k;
- f_k: fraction of total phase-assigned contact signal occurring in phase k;
- b_k: residue-count baseline fraction for phase k;
- e_k: phase enrichment relative to the residue-count baseline;

- $\mathbf{v}$: centered, unit-normalized seven-phase enrichment signature;
- N: number of taxa in a panel;
- R: number of null permutations.

Unless stated otherwise, only residues within the selected contiguous coiled-coil (CC) segment possess finite HARP phase indices and contribute to phase-based calculations.

2 Trajectory-derived contact profile

2.1 Residue-pair contact definition

HARP first converts atomistic MG–MyhT proximity into a residue-pair contact score. In the default `hard` contact mode,

$$w_{tji} = \begin{cases} 1, & d_{tji}^{\min} \le c, \\ 0, & d_{tji}^{\min} > c, \end{cases} \tag{1}$$

where $d_{tji}^{\min}$ is the minimum heavy-atom distance between MG residue j and MyhT residue i in frame t, and c is the contact cutoff. The default cutoff is 4.5 Å. A residue pair therefore scores 1 when any qualifying heavy-atom pair lies within the cutoff. Hydrogen atoms are excluded before the distance calculation, and trajectory box dimensions are supplied to the MDAnalysis distance calculation.

Implementation: `trajectory.py -> compute_contact_trajectory()`.

HARP also implements an optional `smooth` contact mode:

$$w_{tji} = \begin{cases} \dfrac{1}{1 + \left(d_{tji}^{\min}/c\right)^p}, & d_{tji}^{\min} \le 2c, \\ 0, & d_{tji}^{\min} > 2c, \end{cases} \tag{2}$$

where p is `smooth_power`, default $p = 6$. The smooth function is explicitly truncated at $2c$. For qualifying atom pairs, the strongest pairwise weight is retained for each MG–MyhT residue pair.

Implementation: `trajectory.py -> compute_contact_trajectory()`.

2.2 Per-frame MyhT residue signal

HARP reduces the MG $\times$ MyhT residue-pair matrix to one signal for each MyhT residue:

$$x_{ti} = \max_j w_{tji}. \tag{3}$$

A MyhT residue therefore receives the strength of its strongest MG engagement in that frame. Contacts with multiple MG residues are not summed. In hard mode, $x_{ti} \in \{0, 1\}$; in smooth mode, $0 \le x_{ti} \le 1$.

Implementation: `pair_matrix.max(axis=0)` in `trajectory.py`.

2.3 Time-averaged residue contact profile

The profile passed to the taxon-level HARP statistics is

$$x_i = \frac{1}{F}\sum_{t=1}^{F} x_{ti}. \tag{4}$$

In hard mode, x_i is literally the fraction of analysed frames in which MyhT residue i contacts at least one MG residue. In smooth mode, it is the mean smooth contact score. HARP retains the label `contact_occupancy` for both modes, but the probabilistic occupancy interpretation applies literally only to hard mode.

Implementation: `analysis.py` passes `result.myht_frame_signal.mean(axis=0)` to `analyze_profile()`.

For inspection, HARP separately records the MG $\times$ MyhT residue-pair average

$$\bar{w}_{ji} = \frac{1}{F}\sum_{t=1}^{F} w_{tji}. \tag{5}$$

This matrix is written as `mg_myht_pair_occupancy.tsv`. It is not the profile directly used by the phase statistics; those statistics use x_i from Eq. (4).

Implementation: `trajectory.py -> pair_sum / n_frames`.

3 DeepCoil2-anchored HARP register

3.1 Selection of the CC segment

A residue is provisionally CC-positive when

$$CC_i \geq \tau_{CC}, \tag{6}$$

where CC_i is the DeepCoil2 `cc` value and τ_{CC} is the CC threshold, default 0.5. HARP identifies all contiguous CC-positive runs and selects the longest contiguous segment. It does not bridge CC-negative gaps. The DeepCoil2 `raw_cc` column is required in the input but is not used in this calculation.

Implementation: `deepcoil.py -> parse_deepcoil2()`.

3.2 DeepCoil2 a/d anchors

Within the selected CC segment, confident anchor positions are

$$A = \{p_i : P_i(a) \geq \tau_A\}, \qquad D = \{p_i : P_i(d) \geq \tau_A\}, \tag{7}$$

where τ_A is the anchor threshold, default 0.5. At least two confident a calls and two confident d calls are required to establish a register.

Implementation: `deepcoil.py -> parse_deepcoil2()`.

3.3 Dominant modulo-7 frame

For the confident anchor positions, HARP determines the modal sequence-position class modulo seven:

$$r_a = \arg\max_{r \in \{0,\ldots,6\}} N_a(r), \qquad r_d = \arg\max_{r \in \{0,\ldots,6\}} N_d(r). \tag{8}$$

Here $N_a(r)$ and $N_d(r)$ are the numbers of confident a and d calls, respectively, whose sequence positions belong to modulo class r. DeepCoil2 therefore provides the anchor evidence; HARP constructs the complete seven-state coordinate system from that evidence. If modal classes are tied, the numerically smallest winning class is selected and the tie is recorded diagnostically.

Implementation: `deepcoil.py -> _modal_modulo()`.

3.4 Canonical a–d relationship

HARP requires

$$(r_d - r_a) \bmod 7 = 3. \tag{9}$$

Failure of this condition terminates the analysis. The requirement applies to the dominant a and d modulo classes. Individual confident calls may disagree with the dominant frame; those disagreements are retained as register-quality diagnostics.

Implementation: `d_offset = (d_mod - a_mod) % 7`, followed by `d_offset != 3` validation.

3.5 Register purity

Agreement of the anchor calls with their dominant modulo classes is summarized by

$$\pi_a = \frac{N_a(r_a)}{n_a}, \qquad \pi_d = \frac{N_d(r_d)}{n_d}, \tag{10}$$

where $n_a = |A|$ and $n_d = |D|$, and by

$$\pi_{\text{combined}} = \frac{n_a \pi_a + n_d \pi_d}{n_a + n_d}. \tag{11}$$

These quantities diagnose register consistency. They are not used as weights in downstream phase statistics or panel analysis.

Implementation: `_modal_modulo()` and `DeepCoilRegister.register_purity`.

3.6 Assignment of the a–g coordinate

HARP chooses the earliest confident a call belonging to the dominant a modulo class as its a-origin:

$$p_a^{(0)} = \min\{p \in A : p \bmod 7 = r_a\}. \tag{12}$$

Every residue within the selected CC segment is then assigned

$$q_i = (p_i - p_a^{(0)}) \bmod 7. \tag{13}$$

The resulting indices correspond to

$$0, 1, 2, 3, 4, 5, 6 \longleftrightarrow a, b, c, d, e, f, g.$$

Positions outside the selected CC segment receive no finite phase index. Thus, confident DeepCoil2 a/d calls anchor the coordinate system; HARP subsequently propagates that register across the entire selected CC segment.

Implementation: `deepcoil.py -> parse_deepcoil2()`.

4 Phase occupancy and residue-count baseline

4.1 Phase sums and residue representation

For phase k,

$$S_k = \sum_{i:q_i=k} x_i, \qquad n_k = \sum_i I(q_i = k). \tag{14}$$

S_k is the total contact signal assigned to phase k, whereas n_k is the number of valid residues represented in that phase.

Implementation: `stats.py -> phase_sums()`.

4.2 Phase contact fraction

The fraction of all phase-assigned contact signal occurring in phase k is

$$f_k = \frac{S_k}{\sum_{j=0}^{6} S_j}. \tag{15}$$

This is not mean occupancy per residue within the phase. In particular, HARP does not calculate S_k/n_k as its phase contact fraction. If total phase contact signal is zero, all seven fractions are returned as zero.

Implementation: `stats.py -> phase_fraction()`.

4.3 Residue-count baseline

The baseline representation of phase k is

$$b_k = \frac{n_k}{\sum_{j=0}^{6} n_j}. \tag{16}$$

The baseline is therefore determined by the actual residue representation of each phase and is not assumed to be 1/7. This matters particularly at the boundaries of a finite CC segment, where the seven phases need not contain equal numbers of residues.

Implementation: `stats.py -> phase_fraction()`.

4.4 Phase enrichment

Relative enrichment is

$$e_k = \frac{f_k}{b_k}. \tag{17}$$

Thus $e_k = 1$ denotes contact signal proportional to residue representation, $e_k > 1$ relative enrichment, and $e_k < 1$ relative depletion. Where $b_k = 0$, the implementation returns zero rather than dividing by zero.

Implementation: `stats.py -> phase_enrichment()`.

5 Taxon-level statistics

5.1 Sequence-coordinate fundamental period-7 amplitude

HARP reports the descriptive sequence-coordinate amplitude

$$A_{\mathrm{seq},1} = \frac{\left|\sum_i x_i \exp\left(-\frac{2\pi \mathrm{i} p_i}{7}\right)\right|}{\sum_i |x_i|}. \tag{18}$$

This quantity measures the fundamental one-cycle-per-seven-residues component directly in sequence coordinate space. It is not the default inferential statistic `period7_max_mode`.

Implementation: `stats.py -> period7_amplitude()`. In `analysis.py`, the CC mask is supplied as its validity mask.

5.2 Fourier modes of the seven phase bins

After contact signal has been collapsed into the seven registered phases, HARP calculates

$$A_m = \frac{\left|\sum_{k=0}^{6} S_k \exp\left(-\frac{2\pi \mathrm{i} m k}{7}\right)\right|}{\sum_{k=0}^{6} S_k}, \qquad m = 1, 2, 3. \tag{19}$$

These are the three non-redundant non-zero Fourier amplitudes of the seven-bin phase distribution. Modes 2 and 3 are therefore harmonics within the seven-state phase coordinate and should not be interpreted as independent sequence periodicities.

Implementation: `stats.py -> seven_bin_modes()`.

5.3 `period7_max_mode`

The default HARP v4.1 primary statistic is

$$M_7 = \max(A_1, A_2, A_3). \tag{20}$$

The maximization is repeated for every null permutation. Consequently, the permutation probability incorporates selection of the strongest seven-bin harmonic. This statistic should therefore be described precisely as the maximum seven-bin Fourier mode, rather than simply as "the period-7 amplitude."

Implementation: `stats.py -> max_seven_bin_mode()`.

5.4 Phase nonuniformity

Expected contact signal under the residue-count baseline is

$$E_k = \left(\sum_{j=0}^{6} S_j\right) \frac{n_k}{\sum_{j=0}^{6} n_j}. \tag{21}$$

HARP then calculates

$$Q = \sum_{k=0}^{6} \frac{(S_k - E_k)^2}{\max(E_k, 10^{-12})}. \tag{22}$$

This is a Pearson-style concentration statistic, but the S_k values are summed contact signal rather than integer counts. HARP therefore does not use a theoretical χ^2 distribution. Statistical inference comes from the permutation null described below.

Implementation: `stats.py -> phase_nonuniformity()`.

5.5 Target-phase enrichment

For a predefined target set T,

$$E_T = \frac{\sum_{k \in T} f_k}{\sum_{k \in T} b_k}. \tag{23}$$

The default target phases are $T = \{a, d\}$. This is a pooled enrichment ratio, not the mean of the individual enrichment values e_a and e_d.

Implementation: `stats.py -> target_enrichment()`.

6 Taxon-level null model and empirical probability

6.1 Contiguous block shuffle

For the taxon-level null, the CC-selected residue profile is divided sequentially into contiguous blocks $B_1, B_2, \ldots, B_K$. A random permutation π of those blocks produces

$$x^*_{\mathrm{CC}} = B_{\pi(1)} \| B_{\pi(2)} \| \cdots \| B_{\pi(K)}. \tag{24}$$

The default block size is four residues. A terminal block may contain fewer than four residues. Values outside the CC shuffle mask remain fixed. The relevant statistic—`period7_max_mode`, `phase_nonuniformity`, `target_phase_enrichment`, or optional locked-template similarity—is recalculated from every shuffled profile.

Implementation: `stats.py -> block_shuffle()` and `permutation_test()`; `analysis.py` supplies `is_cc_segment` as `shuffle_mask`.

6.2 Plus-one permutation probability

For observed statistic T_{obs} and R null permutations,

$$p = \frac{1 + \sum_{r=1}^{R} I(T^*_r \geq T_{\mathrm{obs}})}{R + 1}. \tag{25}$$

The test is upper-tailed, and ties with the observed statistic count as exceedances. With $R = 9999$, the minimum attainable probability is $1/10000 = 0.0001$. This plus-one correction is used for both taxon-level and panel inference.

Implementation: `stats.py -> _plus_one_p()`.

The null mean and sample standard deviation reported by HARP are

$$\bar{T}^* = \frac{1}{R} \sum_{r=1}^{R} T^*_r, \tag{26}$$

and

$$s_0 = \sqrt{\frac{\sum_{r=1}^{R} (T^*_r - \bar{T}^*)^2}{R - 1}}. \tag{27}$$

HARP additionally records the empirical 95th percentile of the null distribution. These are descriptive summaries; the permutation probability itself is calculated by Eq. (25).

Implementation: `stats.py -> permutation_test()`.

7 Temporal phase structure

7.1 Temporal block profiles

Trajectory frames are divided sequentially into approximately equal-sized frame blocks. For block b,

$$x_i^{(b)} = \frac{1}{F_b} \sum_{t \in b} x_{ti}. \tag{28}$$

The block profile is processed through the same phase calculations as the complete trajectory. HARP uses `np.array_split`, so block sizes differ by at most one frame. If fewer frames than requested blocks are available, the number of blocks is reduced accordingly.

Implementation: `stats.py -> time_block_phase_vectors()`.

7.2 Centered phase signature

For the seven enrichment values,

$$u_k = e_k - \bar{e}, \qquad \bar{e} = \frac{1}{7} \sum_{j=0}^{6} e_j. \tag{29}$$

The centered vector is then Euclidean-normalized:

$$v_k = \frac{u_k}{\sqrt{\sum_{j=0}^{6} u_j^2}}. \tag{30}$$

The resulting seven-component vector $\mathbf{v}$ is the HARP phase signature. Centering removes the common enrichment level, while normalization removes overall Euclidean magnitude. What remains is the relative shape and phase orientation of enrichment across a–g. A zero-norm centered vector remains zero; such a signature is rejected from panel analysis, which explicitly requires unit-norm taxon signatures.

Implementation: `stats.py -> centered_phase_signature()`; panel validation in `analysis.py -> run_panel()`.

7.3 Temporal signature consistency

For B temporal-block signatures,

$$C_{\text{temporal}} = \frac{2}{B(B-1)} \sum_{r<s} \mathbf{v}_r^{\mathsf{T}} \mathbf{v}_s. \tag{31}$$

This is the mean pairwise dot product among the centered, normalized temporal signatures and therefore, for non-degenerate signatures, their mean pairwise cosine similarity. The temporal heatmap and this scalar statistic summarize related but distinct quantities: the heatmap displays block phase contact fractions, whereas `temporal_signature_consistency` operates on block centered enrichment signatures.

Implementation: `stats.py -> temporal_signature_consistency()`.

8 Taxon representation used by the panel

The phase signature passed from each taxon into panel analysis can be written compactly as

$$\mathbf{v} = \frac{\mathbf{e} - \bar{e}\mathbf{1}}{\|\mathbf{e} - \bar{e}\mathbf{1}\|_2}. \tag{32}$$

The panel therefore does not compare raw residue occupancies, total contact signal, phase sums, or raw phase fractions. It compares the shape and a–g orientation of relative phase enrichment after centering and unit normalization. Consequently, absolute contact magnitude is not retained as a direct scale factor in the panel statistic.

Implementation: `stats.py -> centered_phase_signature()`; written to `descriptive.phase_signature` by `analysis.py` and loaded by `run_panel()`.

9 Cross-taxon phase alignment

9.1 Leave-one-out consensus

For taxon i, HARP constructs the consensus of every other taxon:

$$\mathbf{t}_{-i} = \frac{1}{N-1} \sum_{j \neq i} \mathbf{v}_j. \tag{33}$$

This prevents a taxon from contributing to the reference against which it is itself evaluated.

Implementation: local `loo_scores()` inside `stats.py -> panel_leave_one_out_test()`.

9.2 Leave-one-out taxon similarity

Taxon i is compared with its leave-one-out consensus by cosine similarity:

$$s_i = \frac{\mathbf{v}_i^{\mathsf{T}} \mathbf{t}_{-i}}{\|\mathbf{v}_i\|_2 \, \|\mathbf{t}_{-i}\|_2}. \tag{34}$$

This is not mean pairwise taxon similarity. Each taxon is compared once against the mean signature of all other taxa. If either vector has zero norm, the implementation returns zero.

Implementation: `stats.py -> panel_leave_one_out_test() -> loo_scores()`.

9.3 Observed panel statistic

The panel statistic is the arithmetic mean of the leave-one-out similarities:

$$T_{\text{panel}} = \frac{1}{N}\sum_{i=1}^{N} s_i. \tag{35}$$

Thus, the precise mathematical description of the HARP panel statistic is mean leave-one-out consensus phase similarity.

Implementation: `observed = np.mean(observed_taxon)` in `panel_leave_one_out_test()`.

10 Panel null model

10.1 Independent circular rotations

For every panel-null replicate, each taxon independently receives a uniformly sampled circular rotation:

$$r_i \sim \text{Uniform}\{0, 1, \dots, 6\}, \qquad \mathbf{v}_i^* = \text{roll}(\mathbf{v}_i, r_i). \tag{36}$$

Rotation by zero is included. Thus, any individual taxon has probability 1/7 of retaining its observed orientation in a given null replicate. The transformation preserves each taxon's seven-bin shape, centering, norm, and internal phase relationships while disrupting systematic common a–g registration across taxa.

Implementation: `np.roll(row, rng.integers(0,7))` independently for every signature.

10.2 Recalculated panel statistic under the null

For null replicate r,

$$T^*_{\text{panel},r} = \frac{1}{N}\sum_{i=1}^{N} s^*_{i,r}, \tag{37}$$

where the leave-one-out consensus and similarity are recalculated from the rotated signatures. HARP therefore does not compare randomized taxa with the observed consensus. Every null replicate generates its own leave-one-out consensus structure.

Implementation: `null[j] = np.mean(loo_scores(rotated))`.

10.3 Panel permutation probability

The panel probability is

$$p_{\text{panel}} = \frac{1 + \sum_{r=1}^{R} I(T^*_{\text{panel},r} \geq T_{\text{panel}})}{R+1}. \tag{38}$$

This is the same upper-tail plus-one empirical probability used for the taxon-level permutation tests.

Implementation: `panel_leave_one_out_test()` calls `_plus_one_p(observed, null)`.

11 What HARP normalizes, preserves and removes

The transformations above impose several important boundaries on interpretation.

Residue level. HARP reduces simultaneous MG interactions with one MyhT residue to their maximum, rather than summing them. The final taxon profile is the time average of this per-frame maximum.

Phase level. Phase contact signal is normalized by total phase-assigned contact signal to obtain f_k. Residue representation is separately normalized to obtain b_k. Their ratio produces enrichment e_k.

Taxon null. The contiguous-block null rearranges CC-profile values while preserving the values themselves and short-range ordering within each block. Residues outside the shuffle mask remain unchanged.

Panel representation. Phase enrichment is mean-centered and Euclidean-normalized. This removes the common enrichment level and overall vector magnitude before cross-taxon comparison.

Panel null. Circular rotation preserves each taxon's complete seven-phase pattern while randomizing only its orientation relative to the common a–g coordinate.

Register quality. DeepCoil anchor purity and tie information are retained as diagnostics but do not weight the downstream statistical calculations.

12 Statistical interpretation and boundaries

HARP v4.1 contains two conceptually distinct levels of permutation inference.

At the **taxon level**, HARP asks whether the observed phase statistic is unusually strong relative to profiles generated by contiguous block rearrangement of the CC-associated residue contact signal.

At the **panel level**, HARP asks whether the observed common phase registration across taxa is unusually strong relative to panels in which each taxon retains its own phase-pattern shape but receives an independently randomized circular orientation.

These tests therefore address different null hypotheses and their probabilities should not be interpreted interchangeably.

`phase_nonuniformity` has a Pearson-like mathematical form but is not assigned a theoretical chi-square probability. HARP's inferential probabilities are empirical permutation probabilities.

Likewise, the panel statistic does not establish biological homology, evolutionary mechanism, molecular causality, or binding specificity by itself. It quantifies shared alignment of the HARP phase signatures under the implemented statistical model.

13 Mathematical overview of HARP v4.1

The complete statistical architecture can be summarized as follows.

MD trajectory

heavy-atom MG–MyhT distances → residue-pair contact score w_{tji} → strongest MG engagement

$x_{ti} \rightarrow$ time-averaged MyhT profile x_i.

DeepCoil2 register

CC-positive residues $\rightarrow$ longest contiguous CC segment $\rightarrow$ confident a/d anchors $\rightarrow$ dominant modulo-7 classes $\rightarrow$ canonical $d = a + 3$ validation $\rightarrow$ HARP a–g coordinate q_i.

Taxon statistics

$x_i + q_i \rightarrow$ phase sums $S_k \rightarrow$ contact fractions $f_k \rightarrow$ residue-count baseline $b_k \rightarrow$ enrichment $e_k \rightarrow$ Fourier/nonuniformity/target statistics $\rightarrow$ contiguous-block null $\rightarrow$ plus-one empirical probability.

Panel statistics

$e_k \rightarrow$ centering $\rightarrow$ unit normalization $\rightarrow$ taxon phase signature $\mathbf{v} \rightarrow$ leave-one-out consensus $\rightarrow$ taxon cosine similarity $s_i \rightarrow$ mean panel statistic $T_{\text{panel}} \rightarrow$ independent circular taxon rotations $\rightarrow$ recalculated leave-one-out null $\rightarrow$ plus-one panel probability.

This chain constitutes the mathematical machinery implemented by HARP v4.1.

14 Source-code correspondence

`trajectory.py` — heavy-atom filtering, MG–MyhT distance evaluation, hard and smooth residue-pair contact scores, per-frame MyhT residue signal, and residue-pair occupancy matrix.

`deepcoil.py` — CC-segment selection, confident a/d anchors, dominant modulo-7 frame inference, canonical a–d validation, register purity, and complete a–g phase assignment.

`stats.py` — phase sums and fractions, residue-count baseline, enrichment, Fourier statistics, phase nonuniformity, target enrichment, contiguous-block permutation tests, temporal signatures, phase-signature normalization, leave-one-out panel statistic, circular-rotation panel null, and empirical probabilities.

`analysis.py` — integration of trajectory, register, and statistical calculations; selection of the CC shuffle mask; construction of per-taxon outputs; validation and transfer of the seven-component phase signature into panel analysis.

`plotting.py` — graphical representation of quantities calculated upstream. No additional inferential transformation is introduced by the plotting functions.

Closing statement

HARP v4.1 converts trajectory-derived MG–MyhT contact behaviour into a DeepCoil2-anchored seven-phase representation and tests that structure at two levels: within individual taxa and across a multi-taxon panel.

Its taxon-level null tests whether observed phase structure exceeds that expected after constrained rearrangement of the residue contact profile. Its panel-level null asks whether independently phase-randomized taxon signatures reproduce the observed degree of common leave-one-out alignment.

The resulting statistics are therefore conditional on HARP's explicit contact definition, register construction, normalization procedures, and null transformations. This supplement documents those operations so that the numerical outputs of HARP v4.1 can be traced directly from the underlying trajectory-derived contact signal through to the final empirical permutation probabilities.